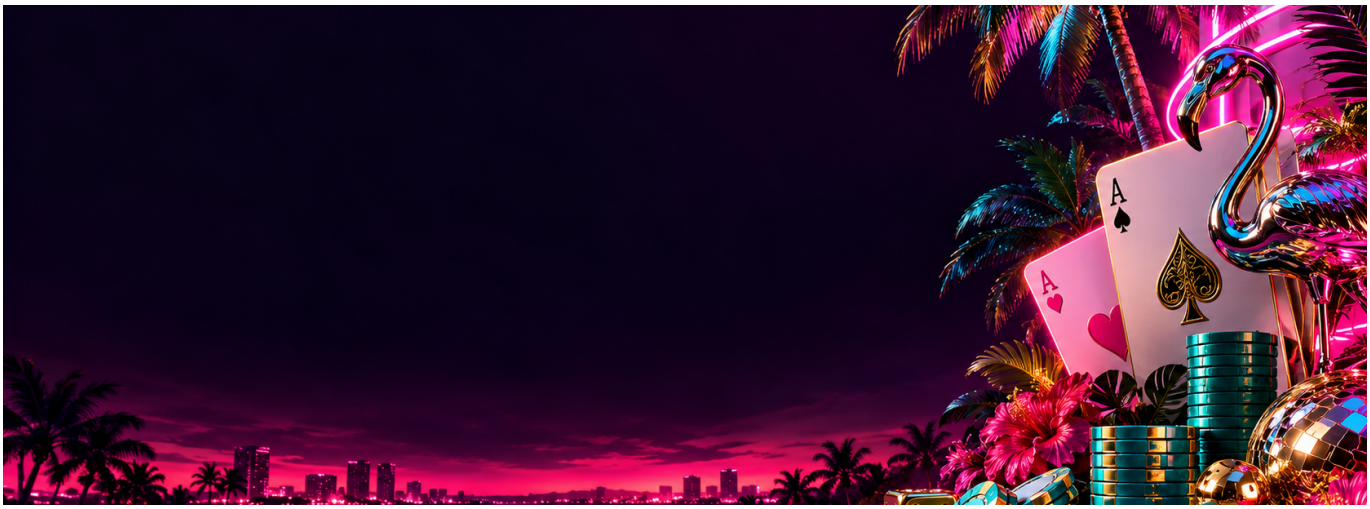
THE CHALLENGE

Blackjack, your way.

Build a casino together. Bring your own twist.



One team. Three connected parts: the blackjack game, the casino engine and the casino website. Use AI to build your own part, then make the whole thing work together.

The experience you are building

Create an account. Add fictional credits. Place a bet. Play blackjack. See the result, your new balance and your transaction history.

Make it yours

Choose one twist together that makes your version more fun or interesting. A coaching mode? A daily challenge? Something nobody else thought of? Your call. Keep it small enough to finish and give each owner a part in making it real.

What this is really for

Show each other how you work with AI. Try a teammate’s technique. Challenge your usual habits. Come away with something you can use on your next project.

YOUR TEAM

Sit together. Own your part.

Share screens and ideas. Keep clear code ownership.

1 / The game

Own the server-side blackjack logic, dealer, deck and playable game UI. Handle Hit, Stand, Double and Split. Track each hand and its stake. Keep hidden cards off the client. Send trusted results to the casino engine.

2 / The casino engine

Own accounts, balances, fictional deposits, bets, payouts and transaction history. Validate every extra stake for splitting or doubling. Store the data and prevent duplicate debits or payouts.

3 / The casino website

Own the lobby, account flow, wallet screens and history. Embed or launch the game owner's UI. Connect the full player journey and refresh balances after bets and payouts. Own clear startup instructions.

Four people? Split an area clearly

Keep the same three components. Divide one component into two named deliveries, such as game logic and game UI. Agree who owns which files. Every person still directs their own AI work and explains their own contribution.

The teams

Team	People
1	Nicolas Villamizar, Mariana, John
2	Edgar, Heriberto, David Portilla
3	Andres L, Nicolas Conde, David R, Luis

HOW YOU WORK

Borrow a better workflow.

AI can do a lot. You still own the result.

The working agreement

• Each person runs their own AI session and owns a clear part of the code. Do not hand the whole casino to one agent.

• Show screens, ask questions and help each other. If another component needs changing, its owner makes the change.

• Agree the interfaces first. Write down IDs, payloads, outcomes and errors. Give shared files a named owner.

• Keep connecting real components as you build. Get one rough player journey working before polishing it.

Try something new

On Friday, each person shows their AI workflow for five minutes. Pick at least one technique from a teammate to try. At Saturday's check-in, show something that worked or failed and what you changed.

A stack that gets you moving

Part	Suggested choice
Code	GitHub Free. One repository per team.
App	TypeScript + Next.js / Node.js. Separate modules, one app is fine.
Database	Neon Free (Postgres). One database per team.
Optional hosting	Vercel Hobby, if its conditions fit your project.

Design, CSS and styling are completely up to you. Use a stack you know if you prefer. Local demos count; nobody needs to buy a plan.

Vercel Hobby is for personal, non-commercial use and has restrictions on private-repository collaboration and GitHub organization repositories. If it does not fit, run locally or use organizer-provided hosting. Keep credentials out of Git and browser code.

BLACKJACK BASICS

Beat the dealer. Stay under 22.

These are our workshop house rules. Use them across all three components.

The deal

One player against the dealer. Shuffle a fresh 52-card deck for each round. Deal two cards each. One dealer card is face up and one stays hidden until the round is resolved. Never deal the same card twice in a round.

What the cards are worth

2 to 10 count at face value. Jack, queen and king count as 10. An ace counts as 11 when it fits without going over 21, otherwise 1. A hand with an ace counted as 11 is a **soft hand**. Ace + 6 is soft 17; add a 10 and it becomes 17 with the ace counting as 1.

Your moves

Move	What happens
Hit	Take another card. You may hit again unless the hand is finished.
Stand	Keep your total and finish this hand.
Double down	Add the same stake again, take exactly one card and finish the hand. See next page.
Split	Turn a starting pair into two separate hands with equal stakes. See next page.

Who wins?

Go over 21 and that hand loses immediately. A hand reaching 21 finishes automatically. Once all player hands are finished, the dealer reveals the hidden card, draws below 17 and stands on all 17, including soft 17. A dealer bust pays every player hand that has not busted. Otherwise, higher total wins; equal totals are a **push** (tie).

Natural blackjack

An ace plus a 10-value card on the original two-card hand is blackjack. Check both sides immediately after the deal, before Split or Double. Both blackjack means a push. Otherwise, blackjack wins immediately. A 21 made after splitting is a normal 21, not natural blackjack.

SPLIT AND DOUBLE

More hands. More at stake.

Both moves need enough credits for an extra stake. Validate before changing anything.

Double down

On any two-card hand, before taking a Hit, you may add a stake equal to that hand's current stake. You receive exactly one more card, then the hand ends. You cannot Hit, Split or Double again on that hand. Doubling after a split is allowed, except on split aces.

Example: You bet 10 and receive 5 + 6. Double adds another 10, making this hand's stake 20. You take one card. If you win, the total return is 40 credits; a push returns 20; a loss returns zero.

Split a pair

If your original two cards have the **same rank**, you may split them. Two 8s qualify; a king and queen do not. Add one extra stake equal to the original stake. Put one original card in each hand, then deal one new card to each, left hand first.

Play the left hand first, then the right. Each hand has its own total, available actions, stake and result. The dealer plays once after both hands finish. Settle each hand separately against that same dealer hand.

The limits for this workshop

• One split per round, for a maximum of two player hands. No re-splitting.

• Non-ace split hands can Hit or Stand as usual, and Double before taking any Hit.

• Split aces receive exactly one new card each, then both hands stand. No Hit or Double.

• Ace + a 10-value card after a split counts as 21 and pays as a normal win.

• No insurance, surrender or side bets. Only fictional credits.

Split example

You bet 10 and receive 8 + 8. Split costs 10 more and creates two 10-credit hands. If one wins and one loses, the combined return is 20. If instead you Double the left hand before taking a Hit, add 10 more and make that hand's stake 20. The right hand's stake stays 10.

CREDITS AND CHECKS

Make the numbers add up.

One result per hand. One payout per result. Retries must be safe.

Our simple payout rule

Use integer credits. New accounts start at zero. Deposits and stakes must be positive whole numbers. Debit the stake when accepting it, including extra stakes for Split and Double.

Result	Total returned to the wallet
Win	2 times that hand’s final stake.
Push	That hand’s final stake.
Loss	Zero.
Natural blackjack	Same as a win: 2 times the stake for this workshop.

These are deliberately simplified even-money payouts, including natural blackjack. With 100 credits and a 10-credit bet, finish at 110 after a win, 100 after a push or 90 after a loss.

Keep state and money in sync

The server decides cards and results. The casino engine owns all balance changes. Agree account, round, hand, bet and transaction IDs. A round can now contain two hands. Record original stakes, extra stakes and each hand’s payout so the history tells the full story.

Check and debit extra funds atomically before accepting Split or Double. If the wallet has insufficient funds, reject the move without changing cards, hands or balances. Repeating a request must never draw another card, debit twice or pay twice. Persist the data needed to finish and settle a round reliably.

Run these checks before the demo

- Core flow: account, deposit, bet, play, result, balance and history.
- Rules: flexible aces, natural blackjack, bust, push and dealer soft 17.
- Double: one extra stake, exactly one card, then no more actions.
- Split: two hands, correct stakes and results, no re-split; split aces get one card.
- Money: insufficient funds leaves state unchanged; retries do not debit or pay again.
- Your twist: one visible success criterion that you can demonstrate.

FRIDAY AND SATURDAY

Plan Friday. Build Saturday.

Three hours of building, then demos and shared learning.

Friday / introduction within 15:45-17:00

This slot also includes AI Champs news. The facilitators confirm the split. Before leaving, assign owners, show your workflows, choose a small twist, agree interfaces and check repository, database and local startup access.

Saturday / the build

Time	What happens
09:00-09:30	Breakfast
09:30-10:50	Build 1. Start immediately. Connect a rough account, wallet and game journey.
10:50-11:00	Break and workflow exchange. Show a success or failure and resolve blockers.
11:00-11:55	Build 2. Complete the rules, checks and a small version of your twist.
11:55-12:10	Peer playtest. Another team plays and reports an issue.
12:10-12:30	Critical fixes, rerun checks and prepare the demo.
12:30-13:15	Lunch
13:15-14:30	Demos, shared learning and winners
14:30 onwards	Free Time / Team Activity (TBD)

Your five-minute demo

2 minutes: show the full journey, a wallet transaction and your twist.
2 minutes: everyone explains their own delivery and a decision they made.
1 minute: share an AI mistake you caught and a technique you borrowed.

Leave us with

Runnable code. A clear startup command. Checks we can run. Known limitations. And one thing each of you will use again. For the four-person team, everyone still gets a voice in the demo.